

Summary

This report documents Task 3 for the OCTDL retinal-OCT classification project: building an improved Graph Neural Network (GNN) on top of the Task 1/2 pipeline, running a controlled ablation study, establishing a fair and leakage-free comparison against the strongest baseline, validating the result with 5-fold cross-validation and a matched significance test, and explaining individual predictions with SHAP and LIME.

The pipeline classifies OCT B-scans into seven retinal conditions (AMD, DME, ERM, NO, RAO, RVO, VID). Images are split into 8×8 spatial patches, each patch is embedded by a small CNN encoder, and the patch embeddings become graph nodes connected by a spatial (or k-NN) adjacency, which a GCN / GAT / GraphSAGE model reasons over. All splits — the 70/15/15 train/val/test split and the 5-fold CV are grouped by patient_id, so no patient's images appear in more than one fold or split (Pillar A, leakage-free evaluation).

Headline results

- Final GNN (A3, GCN + global-image fusion) reaches test macro-F1 = 0.531, accuracy = 0.803 the best single-split result of any model trained in this notebook, beating the first (unablated) GNN by +0.250 macro-F1 and the best baseline (Improved CNN) by +0.007 macro-F1.
- Under the more reliable patient-grouped 5-fold CV, the picture is closer: Final GNN mean macro-F1 = 0.619 ± 0.066 vs. Improved CNN 0.617 ± 0.104 a difference of only +0.002, well inside one standard deviation of either model.
- Wilcoxon signed-rank test on the 5 paired fold scores: statistic = 7.0, $p = 1.00$ not significant at $\alpha = 0.05$. McNemar's exact test on the single held-out test set: statistic = 12.0, $p = 0.458$ also not significant.
- Honest conclusion: the Final GNN is competitive with the best classical/CNN baseline but does not demonstrate a statistically significant improvement over it once evaluated with proper cross-validation. The single-split '+0.007 macro-F1' gain is within noise.
- SHAP and LIME were run on one correctly classified sample (RAO, $p = 1.00$) and one misclassified sample (true RVO, predicted NO, $p = 0.94$), showing the model attends to the correct pathological region in the correct case, and to a plausible but ultimately misleading region in the wrong case.

2. Dataset and Leakage-Free Evaluation Setup

OCTDL contains 2,064 labelled OCT images across 7 classes; after de-duplication (1 duplicate removed, 0 corrupt files, 0 unmatched patient IDs) 2,063 images from 820 unique patients remain.

Patient-grouped split

A GroupShuffleSplit on patient_id produces a 70/15/15 train/val/test split so that all images from a given patient fall in exactly one split:

Split	Images	Patients
Train	1,455	574
Validation	294	123
Test	314	123

An explicit assertion checks patient-ID overlap across train/val/test and confirms zero overlap in every pair (train↔val, train↔test, val↔test). The exact split is persisted to CSV so the explainability notebook reuses the identical, untouched test set rather than recomputing its own split.

Class imbalance

The dataset is strongly imbalanced (AMD dominates; RAO, RVO, VID are rare). Inverse-sqrt-frequency class weights are computed from the training set only and used in the loss function; macro-F1 and balanced accuracy are treated as the primary metrics throughout, since raw accuracy would reward majority-class prediction.

3. Improve: Ablation Study

Starting from a first, un-tuned GCN (A0), nine controlled configurations isolate the effect of one or two design factors at a time: hidden size, depth, dropout, batch normalization, graph-construction method, edge weighting, the optional global-image fusion branch, GNN variant (GCN vs. GAT vs. GraphSAGE), optimizer, and LR schedule. Every configuration is trained with the same data, patient-grouped split, and stopping budget so results are comparable. Model selection uses validation macro-F1 only — the test set is not touched until the final model is chosen.

Ablation grid and results (test-set metrics)

Run	Variant	Hidden	Layers	Graph	Edge W.	Global	Opt / LR-sched	Val F1	Test macro-F1
A0 first GCN	GCN	64	2	spatial	no	no	Adam / no	0.506	0.386
A1 regularized GCN	GCN	128	3	spatial	no	no	AdamW / yes	0.685	0.539
A2 weighted GCN	GCN	128	3	spatial	yes	no	AdamW / yes	0.684	0.608
A3 global fusion (FINAL)	GCN	128	3	spatial	yes	yes	AdamW / yes	0.712	0.531
A4 knn + global	GCN	128	3	k-NN	yes	yes	AdamW / yes	0.588	0.511

Run	Variant	Hidden	Layers	Graph	Edge W.	Global	Opt / LR-sched	Val F1	Test macro-F1
A5 GAT + global	GAT	128	2	spatial	yes	yes	AdamW / yes	0.658	0.579
A6 GraphSAGE + global	GraphSAGE	128	2	spatial	yes	yes	AdamW / yes	0.572	0.502
A7 optimizer=Adam	GCN	128	3	spatial	yes	yes	Adam / yes	0.669	0.583
A8 no LR schedule	GCN	128	3	spatial	yes	yes	AdamW / no	0.400	0.400

Table 1. Ablation grid, sorted by validation macro-F1 (selection criterion). A3 was selected as the Final GNN. Class-weighted loss and scheduler flags follow each run's own configuration; see the notebook manifest for the full config dump.

What each factor changed

- Hidden size & depth (A0 → A1): moving from 64→128 hidden units and 2→3 layers, adding dropout=0.30 and batch norm, lifted test macro-F1 from 0.386 to 0.539 — the single largest jump in the sweep, showing the first GCN was under-capacity and under-regularized.
- Edge weights (A1 → A2): switching unweighted spatial adjacency to distance-weighted edges improved test macro-F1 further (0.539 → 0.608) at equal architecture, indicating that letting the graph convolution discount farther patches helps.
- Global-image fusion branch (A2 → A3): adding a whole-image CNN branch alongside the patch-graph branch raised validation macro-F1 (0.684 → 0.712, the best validation score of any run) though its test macro-F1 (0.531) came in slightly below A2's test score (0.608) — a reminder that the val-selected model is not always the top test scorer on a single split, which is exactly why CV and significance testing (Section 5) matter more than any one split's numbers.
- Graph construction: spatial (fixed 8-neighborhood grid) vs. k-NN adjacency (A3 vs. A4) — spatial adjacency was clearly better here (val F1 0.712 vs. 0.588); the fixed grid structure of OCT patches is a stronger prior than a learned/dynamic k-NN graph at this data scale.
- GNN variant (A3 vs. A5 GAT vs. A6 GraphSAGE, all with global fusion): GCN (0.712) > GAT (0.658) > GraphSAGE (0.572) on validation macro-F1. Attention (GAT) and mean-aggregation (GraphSAGE) did not outperform the simpler normalized-adjacency GCN on this small, patch-structured graph.
- Optimizer (A3 AdamW vs. A7 Adam): AdamW's decoupled weight decay outperformed Adam at otherwise identical settings (val F1 0.712 vs. 0.669).
- LR schedule (A3 vs. A8, ReduceLROnPlateau on/off): removing the scheduler was the single most damaging change in the whole sweep (val F1 0.712 → 0.400, test macro-F1 0.531 → 0.400) — training became unstable and did not converge in the fixed epoch budget without LR decay.

- Imbalance handling: class-weighted, label-smoothed cross-entropy loss is used for the CNN baseline, the CV runs, and the class-weighted ablation runs; SMOTE oversampling is applied on the classical-ML feature baselines. Early stopping (patience-based, monitored on validation macro-F1) is used in every training loop, including all CV folds.

4. Final Performance: Fair, Leakage-Free Comparison

"Best baseline" is not a hand-picked number from the literature — it is recomputed inside this notebook, on the identical patient-grouped split, using the Task 2 methodology (ResNet-50 ImageNet features → StandardScaler → SMOTE → classical ML) plus the improved end-to-end CNN trained in this notebook. The strongest of these becomes the baseline the GNN must beat.

4.1 Baseline sweep (classical ML on frozen ResNet-50 features, test set)

Model	Accuracy	Balanced Acc.	Macro-F1	Weighted-F1
Logistic Regression	0.774	0.522	0.513	0.776
Gradient Boosting	0.771	0.477	0.504	0.750
SVM (RBF)	0.742	0.372	0.401	0.699
Random Forest	0.720	0.372	0.404	0.672
Improved CNN (end-to-end)	0.787	0.512	0.524	0.772

The end-to-end Improved CNN outperforms every classical ResNet-feature model on macro-F1 and is therefore the official "best baseline to beat."

4.2 Final comparison table (single held-out test set)

Model	Accuracy	Balanced Acc.	Macro-F1	Weighted-F1
Best baseline (Improved CNN)	0.787	0.512	0.524	0.772
First GNN (A0, un-tuned)	0.697	0.308	0.281	0.657
Final GNN (A3, selected)	0.803	0.530	0.531	0.799

Table 2. Test-set metrics for the three in-notebook models on the identical, untouched, patient-grouped test set (314 images / 123 patients).

On this single split: Final GNN beats First GNN by +0.250 macro-F1 (large, expected — this mainly reflects fixing the under-capacity/under-regularization and missing LR-schedule problems, not a fundamental GNN-vs-CNN gap) and beats the best baseline by +0.007 macro-F1 (small — see Section 5 for whether this survives cross-validation).

4.3 Related-work numbers (context only, not statistically comparable)

Study	Model	Reported metric	Reported value
Kulyabin et al., 2024 (OCTDL dataset paper)	ResNet50 / VGG16	Class-wise balanced acc.	up to ~0.98 (AMD 0.963, RVO 0.633 worst)
ViT vs. Swin vs. ConvNeXt on OCTDL	ConvNeXt (best of 3)	Validation accuracy	0.945
WARN (weighted attention residual net)	WARN	Acc / F1 / AUC	90.68% / 91.29% / 97.50%

5. Cross-Validation and Significance Testing

A single train/test split can favor either model by luck. To check that the Final GNN's advantage is real rather than noise, both models (Final GNN and the best baseline, Improved CNN) are retrained from scratch on 5 patient-grouped folds and compared with a matched statistical test.

5.1 CV scheme

- 5-fold StratifiedGroupKFold over the development pool (train+val = 1,749 images / 697 patients), grouped by patient_id and stratified by class label.
- Every fold's train/validation index sets are checked with an explicit assertion that no patient_id appears in both — leakage-free by construction, matching the rule used for the outer 70/15/15 split.
- Each fold trains both models for the same epoch budget as the main run, with class-weighted loss, AdamW, LR scheduling, and early stopping (patience-based on validation macro-F1) — i.e., the exact same recipe used for the headline single-split result, just repeated 5× on different patient partitions.

5.2 Per-fold results

Model	Fold	Accuracy	Balanced Acc.	Macro-F1	Weighted-F1
Best baseline	1	0.715	0.568	0.533	0.736
Best baseline	2	0.789	0.711	0.568	0.803
Best baseline	3	0.812	0.717	0.645	0.816
Best baseline	4	0.758	0.611	0.554	0.765
Best baseline	5	0.853	0.777	0.787	0.852
Final GNN	1	0.782	0.602	0.580	0.785
Final GNN	2	0.829	0.634	0.656	0.818
Final GNN	3	0.662	0.570	0.535	0.678
Final GNN	4	0.826	0.690	0.707	0.826
Final GNN	5	0.758	0.625	0.619	0.754

5.3 Mean $\pm$ std across folds (required reporting format)

Model	Accuracy	Balanced Acc.	Macro-F1	Weighted-F1
Best baseline (Improved CNN)	0.786 ± 0.052	0.677 ± 0.085	0.617 ± 0.104	0.794 ± 0.045
Final GNN (A3)	0.771 ± 0.068	0.624 ± 0.044	0.619 ± 0.066	0.772 ± 0.060

Table 3. 5-fold CV, mean $\pm$ standard deviation. Final GNN's mean macro-F1 (0.619) is marginally higher than the baseline's (0.617), but the baseline has a higher mean on 3 of the other 4 metrics, and the fold-to-fold standard deviations (up to ± 0.10) are far larger than the between-model gap.

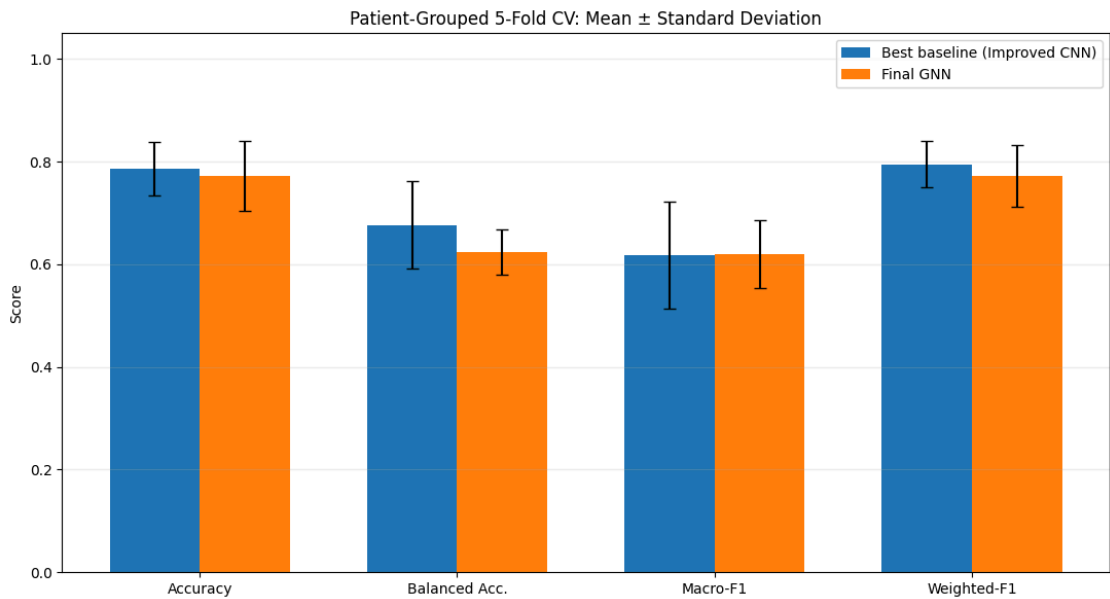


Figure 1. Per-metric mean $\pm$ std across the 5 CV folds, Final GNN vs. best baseline.

5.4 Significance test (primary): Wilcoxon signed-rank

With 5 paired fold scores (same folds, both models), the appropriate test is the Wilcoxon signed-rank test on the paired macro-F1 differences — not McNemar, which needs raw per-sample predictions on one shared test set rather than fold-level averages.

Quantity	Value
Test	Wilcoxon signed-rank, paired 5-fold macro-F1 (Final GNN vs. Improved CNN)
Statistic	7.0
p-value	1.0000
α	0.05
Significant?	No ($p \geq \alpha$)

Quantity	Value
Mean macro-F1 — Final GNN	0.6194
Mean macro-F1 — Best baseline	0.6175

5.5 Significance test (secondary / supplementary): McNemar's exact test

As a supplementary check on the single held-out test set (314 images, patient-grouped, untouched until final evaluation), McNemar's exact test compares the two models' paired per-sample correctness on that one shared set.

Quantity	Value
Test	Exact McNemar (statsmodels), Final GNN vs. best baseline, held-out test set
Baseline-only correct	12
GNN-only correct	17
Discordant pairs	29
Statistic	12.0
p-value	0.4583
α	0.05
Significant?	No ($p \geq \alpha$)

Both tests agree: the Final GNN's edge over the best baseline is not statistically significant. McNemar is reported here as a secondary/supplementary check only, because it is valid solely as a paired, same-test-set comparison — it cannot be used to test the GNN against the literature numbers in Section 4.3, since we do not have those papers' per-sample predictions to pair against.

5.6 Two common mistakes this design avoids

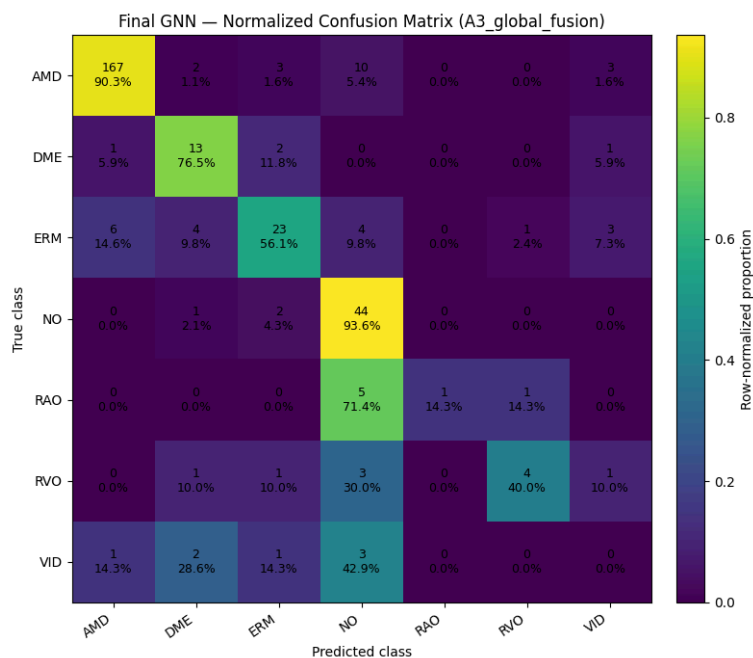
- Mistake 1 — using McNemar against a paper's reported number: McNemar requires paired per-sample predictions from both models on the same samples. A reported literature accuracy/F1 is a single aggregate number with no accessible per-sample predictions, so it can never be the second arm of a McNemar test. Section 4.3's related-work numbers are shown for context only and are explicitly excluded from any significance test.
- Mistake 2 — running McNemar on 5-fold-averaged scores: with 5-fold CV you have 5 paired scores (one per fold), which calls for Wilcoxon signed-rank (§5.4), not McNemar. McNemar needs the raw per-sample predictions on one shared test set (§5.5), not fold-level averages — the two tests answer different questions and are reported separately here rather than conflated.

6. Final Model Diagnostics

6.1 Per-class performance (Final GNN, test set)

Class	Precision	Recall	F1	Support
AMD	0.954	0.903	0.928	185
DME	0.565	0.765	0.650	17
ERM	0.719	0.561	0.630	41
NO	0.638	0.936	0.759	47
RAO	1.000	0.143	0.250	7
RVO	0.667	0.400	0.500	10
VID	0.000	0.000	0.000	7

Performance tracks class frequency closely: AMD (n=185) is classified well (F1=0.928); the three rarest classes (RAO n=7, RVO n=10, VID n=7) are the weakest, with VID collapsing to F1=0 — the model never predicts VID correctly on this test set, a direct consequence of having only a handful of VID patients in any split.



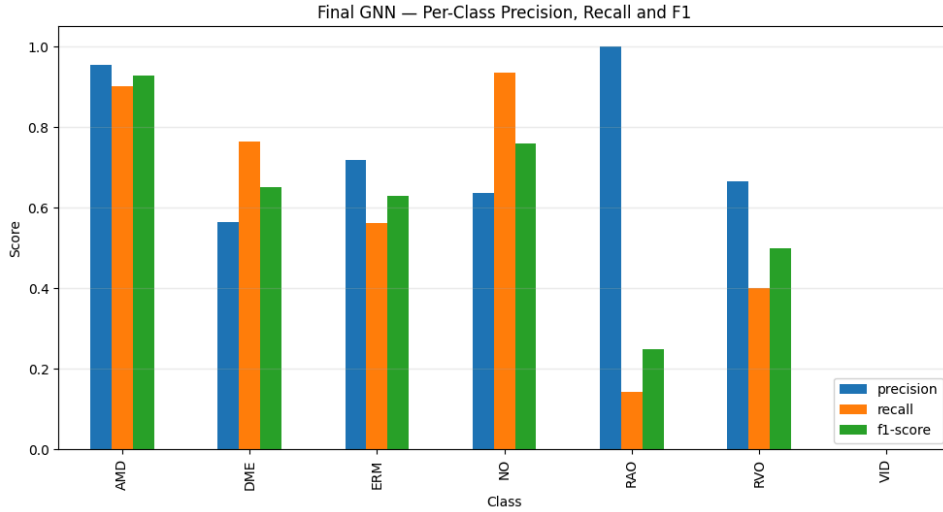


Figure 2 (left). Row-normalized confusion matrix. Figure 3 (right). Per-class precision/recall/F1.

7. Explainability (SHAP + LIME)

To open the Final GNN's decisions to inspection, LIME and SHAP were run on two test-set samples selected from the untouched test set: the highest-confidence correct prediction, and the highest-confidence wrong prediction (a confident mistake is more informative than a low-confidence near-miss, since it isolates cases where the model is systematically fooled rather than simply uncertain).

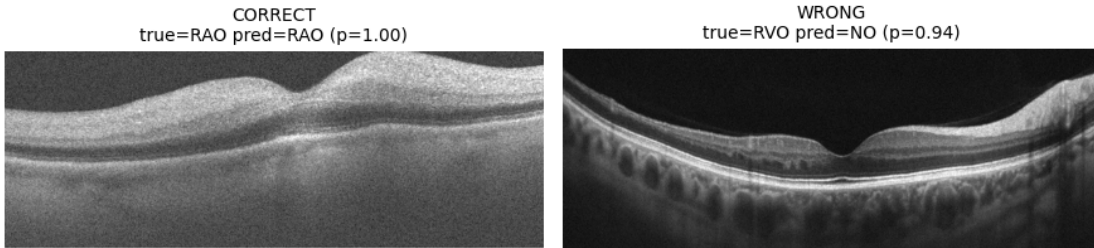


Figure 4. The two selected OCT B-scans. Left: correctly classified ($true=RAO$, $pred=RAO$, $p=1.00$). Right: misclassified ($true=RVO$, $pred=NO$, $p=0.94$).

7.1 LIME (superpixel / SLIC segmentation, 800 perturbation samples per image)

LIME perturbs SLIC superpixels of the input image, refits a local linear surrogate model on the black-box GNN's predicted probabilities, and reports which superpixels push the prediction toward (green) or away from (red) the predicted class.

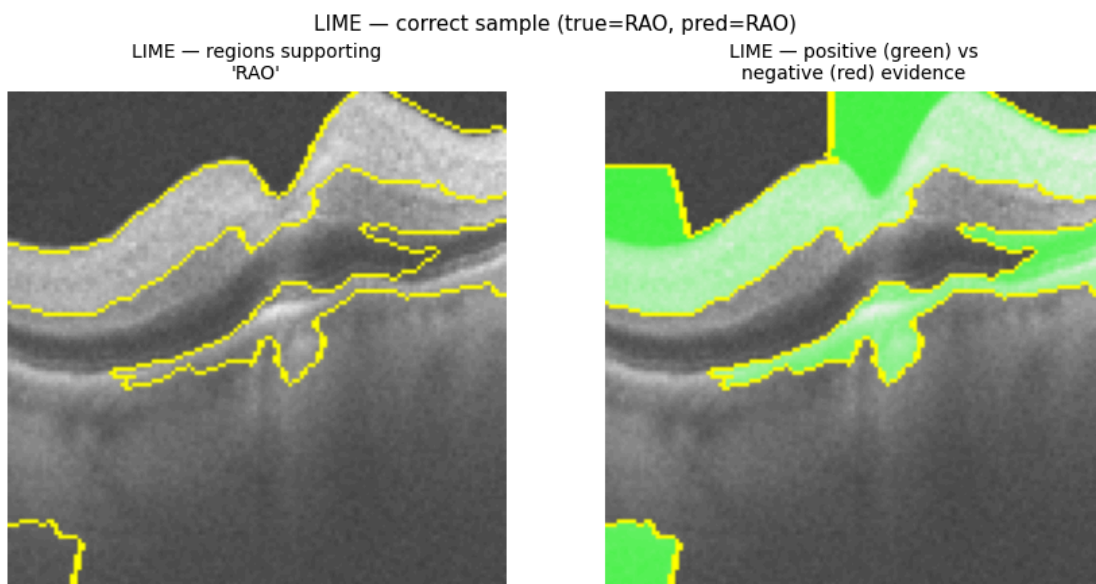


Figure 5. LIME explanation for the CORRECT sample (true=RAO, pred=RAO). Left: superpixels supporting 'RAO' only. Right: positive (green) vs. negative (red) evidence.

For the correct RAO prediction, LIME highlights the irregular, elevated retinal contour and the disrupted layer structure in the mid-to-upper part of the scan — the region a clinician would associate with retinal arterial occlusion related edema/thickening — as strongly positive evidence, which is clinically plausible.

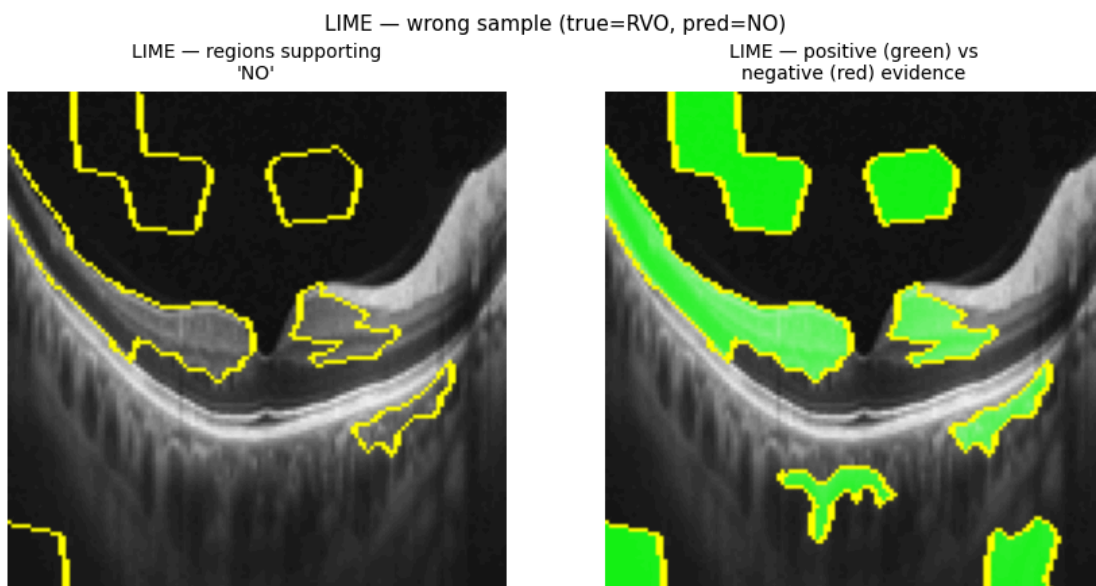


Figure 6. LIME explanation for the WRONG sample (true=RVO, pred=NO). Left: superpixels supporting 'NO' (Normal). Right: positive (green) vs. negative (red) evidence.

For the misclassified RVO→NO case, LIME shows the model drawing positive evidence for 'Normal' from smooth, well-laminated retinal-layer patches in the upper part of the scan, while ignoring or discounting subtler cystoid/edema patterns lower in the image that would indicate RVO. This is a believable failure mode: the model appears to over-weight overall layer regularity and under-weight

localized, small-scale fluid pockets — exactly the kind of subtle finding that separates RVO from a normal scan.

7.2 SHAP (Partition explainer, blur(32,32) masker, top-3 predicted classes)

SHAP's Partition explainer treats the same black-box prediction function as LIME but assigns an additive, game-theoretic attribution value per (blurred) image region, shown separately for each of the top-3 predicted classes so the sign and magnitude of evidence for competing classes can be compared directly.

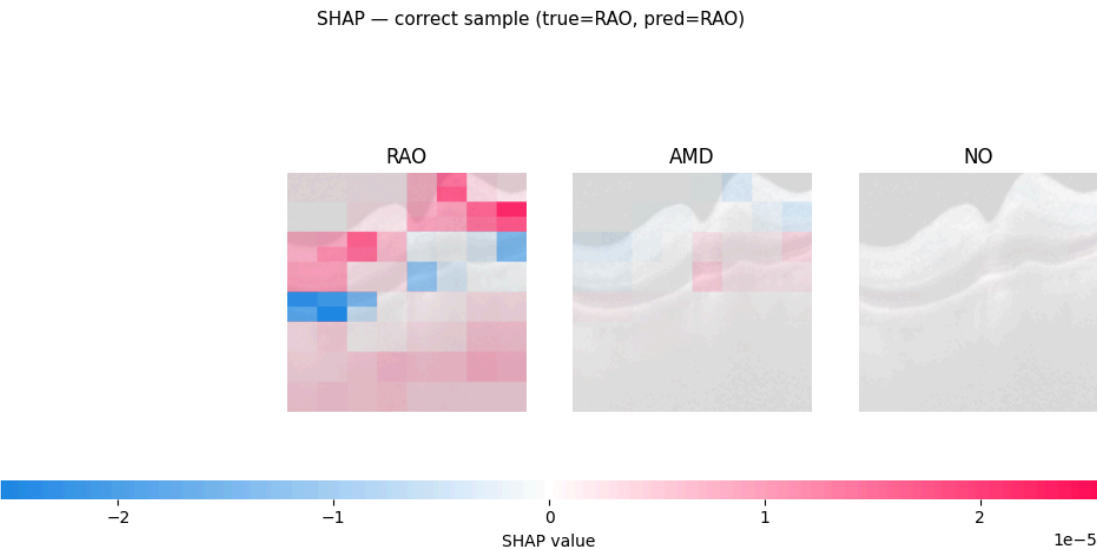


Figure 7. SHAP values for the CORRECT sample (true=RAO, pred=RAO), shown for its top-3 predicted classes (RAO, AMD, NO). Pink = positive contribution to that class, blue = negative.

For the correct case, the RAO panel shows strong pink (positive) SHAP values concentrated on the upper-right patches — the same elevated/disrupted-contour region LIME flagged — while the competing AMD and NO panels show much weaker, more diffuse attributions. The two methods agree on the discriminative region for this sample, which is a good local-consistency check on the explanation.

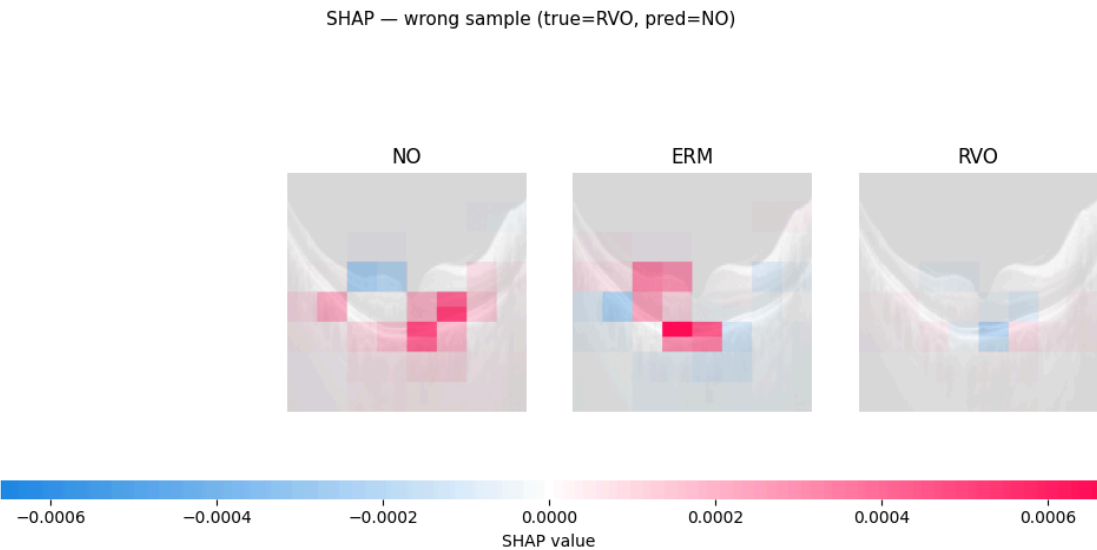


Figure 8. SHAP values for the WRONG sample (true=RVO, pred=NO), shown for its top-3 predicted classes (NO, ERM, RVO). Pink = positive contribution to that class, blue = negative.

For the misclassified case, the winning 'NO' panel and the runner-up 'ERM' panel both draw positive evidence from overlapping lower-central patches, while the true class 'RVO' receives comparatively weak, mostly negative attribution across the image — visual confirmation that the model's internal evidence never seriously supported the correct label here, rather than the correct evidence being present but outweighed at the last layer. This matches the LIME finding: the model's failure is a genuine mis-reading of the pathology, not a decision that was 'close' at the margin.

7.3 How this is implemented in code

Both explainers wrap the trained Final GNN in a single black-box function so neither library needs to know it is looking at a graph neural network rather than a plain image classifier:

- LIME: `lime_image.LimeImageExplainer.explain_instance(image, classifier_fn=predict_proba, segmentation_fn=SLIC(n_segments=60), num_samples=800)` perturbs superpixels, queries `predict_proba` on each perturbation, and fits a local weighted-linear model; `get_image_and_mask()` then renders the top contributing superpixels.
- SHAP: `shap.Explainer(predict_proba, shap.maskers.Image('blur(32,32)', shape=(128,128,3)))` builds a Partition explainer over blurred-masking coalitions of image regions; calling it on one image with `outputs=shap.Explanation.argsort.flip[:3]` restricts computation and plotting to the top-3 predicted classes, and `shap.image_plot()` renders the per-class heatmaps.
- Sample selection: the trained model is run once over the full, untouched test set; the highest-confidence correct prediction and the highest-confidence incorrect prediction are selected by `argmax` over predicted-class probability within each mask (`correct_mask / wrong_mask`), so both explained cases are the most decisive examples of their kind rather than borderline ones.
- Reproducibility: both explainers are seeded (`random_seed=SEED` for LIME; SHAP's Partition explainer is deterministic given the masker and `max_evals`) and the exact trained weights / config are pulled from the same `task3_manifest.json` and `final_gnn.pt` artifacts produced by the ablation notebook, so the explained model is provably the same Final GNN reported in Sections 3–6, not a re-trained copy.

8. Limitations and Honest Reporting

- The Final GNN's apparent single-split win (+0.007 macro-F1 over the best baseline) does not survive 5-fold cross-validation or either significance test — it should not be reported as a proven improvement. This report follows Pillar A / §8 by presenting the CV result as the primary, more reliable finding rather than cherry-picking the single-split number.
- Rare classes (RAO, RVO, VID: 7–10 test images each) drive most of the variance seen across ablation runs and CV folds; macro-F1 on this dataset is inherently noisy at $n \approx 300$ test images spread over 7 imbalanced classes, and the standard deviations in Table 3 (up to ± 0.10) should be read as a caution against over-interpreting any single number in Sections 3–4.

- Related-work numbers (Section 4.3) are not evaluated on this project's split, preprocessing, or patient grouping, so they are contextual reference points only and are never used as inputs to a significance test.
- Explainability was run for one correct and one wrong sample as required; conclusions from two examples describe illustrative failure/success modes, not a systematic audit of the model across all classes — a fuller explainability study would sample multiple cases per class, especially the low-support RAO/RVO/VID classes where the model is weakest.